Exercise 08: Inferring FFs

- Below is the implementation of a D-latch

```
module latch(d,clk,q);
  input d, clk;
  output reg q;

  always @(clk)
    if (clk)
      q <= d;

endmodule
```

Is this code correct?
If so why does it correctly infer and model a latch?
If not, what is wrong with it?

- Submit a single file called **HW2_prob5.sv** containing the following:
  1. In the comments, answer the question about the latch posed above.
  2. Model a D-FF with an active high synchronous reset.
  3. Model a D-FF with asynchronous active low reset and an active high enable.
  4. Model an SR FF with active high synchronous reset, and an active high synchronous set. SR meaning it has an S input that will set the flop, and an R input that will reset the flop. It maintains state if neither S or R are high. If both S and R were asserted R would have priority. It also has active low async reset. This is a handy style flop that we will use frequently.
  5. Answer the following in comments: Does the use of the always_ff construct ensure the logic will infer a flop? Looking for a little more than a simple yes/no answer.

You should use **always_ff** in your design above and have separate always blocks for each.